

LE HAVRE-OCTEVILLE AP, France

WMO: 070460

Lat: 49.5331N

Lon: 0.0917E

Elev: 94

StdP: 100.20

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.2	-1.7	-7.0	2.1	-1.2	-5.2	2.5	-0.7	22.7	8.0	19.1	9.2	4.0	70	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	27.2	19.3	24.7	18.1	22.7	17.2	20.2	25.5	19.0	22.9	18.1	21.4	4.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.4	21.8	17.5	12.7	20.7	16.8	12.1	19.6	58.2	25.7	54.4	23.0	51.5	21.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	11.6	10.0	DB	-4.9	31.9	2.1	3.0	-6.5	34.1	-7.7	35.9	-8.9	37.6	-10.5	39.8	(4)
(5)			WB	-5.6	22.0	2.3	1.2	-7.3	22.8	-8.6	23.5	-9.9	24.1	-11.5	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	5.5	5.5	7.7	10.1	12.8	15.9	17.7	17.7	15.9	12.9	9.1	6.4	(6)	
	DBStd	5.33	3.50	3.38	3.07	3.09	2.90	2.72	2.36	2.11	2.30	2.95	2.85	3.41	(7)	
	HDD10.0	566	140	128	84	35	5	0	0	0	0	9	48	117	(8)	
	HDD18.3	2564	397	359	330	248	174	85	40	36	79	168	276	371	(9)	
	CDD10.0	1102	2	2	12	37	91	176	238	240	177	99	22	5	(10)	
	CDD18.3	59	0	0	0	0	3	11	21	18	6	1	0	0	(11)	
	CDH23.3	379	0	0	0	3	19	92	137	95	30	4	0	0	(12)	
	CDH26.7	100	0	0	0	0	2	25	41	27	6	1	0	0	(13)	
(14) Wind	WSAvg	5.0	5.8	5.6	5.3	4.6	4.6	4.4	4.4	4.4	4.4	4.9	5.6	5.8	(14)	
(15) Precipitation	PrecAvg	852	74	62	53	56	59	63	57	69	70	95	96	100	(15)	
	PrecMax	1125	157	120	180	117	106	142	124	176	159	207	221	205	(16)	
	PrecMin	532	8	12	10	5	12	9	9	9	15	11	23	22	(17)	
	PrecStd	125	36	32	35	32	25	35	29	43	34	48	45	43	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	14.7	18.8	23.1	25.7	29.9	31.0	30.1	27.0	22.6	16.8	13.7	(19)	
		MCWB	11.2	9.7	12.2	14.4	17.6	20.4	20.9	20.5	18.3	16.9	14.7	11.7	(20)	
	2%	DB	11.8	12.2	16.0	19.8	22.6	26.0	26.8	25.9	23.4	19.8	15.0	12.6	(21)	
		MCWB	10.5	9.6	11.2	13.6	16.1	19.0	19.5	18.9	17.2	16.4	13.0	11.1	(22)	
	5%	DB	10.7	11.0	13.7	17.1	20.0	22.9	24.1	23.1	21.2	17.9	13.9	11.7	(23)	
		MCWB	9.6	8.9	10.3	12.2	14.8	17.4	18.1	17.9	16.5	15.4	12.3	10.5	(24)	
	10%	DB	9.9	9.9	11.9	14.9	17.7	20.6	21.8	21.1	19.6	16.8	13.0	10.8	(25)	
		MCWB	8.8	8.2	9.5	11.1	13.9	16.6	17.2	17.2	16.1	14.6	11.5	9.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.1	13.1	15.8	19.1	21.3	21.6	21.4	20.1	17.8	15.4	12.4	(27)	
		MCDB	12.2	12.6	17.2	21.0	24.2	28.0	28.8	28.0	23.9	20.3	16.4	13.1	(28)	
	2%	WB	10.8	10.2	11.8	13.8	16.8	19.4	20.1	19.7	18.4	16.7	13.5	11.5	(29)	
		MCDB	11.5	11.7	14.5	18.5	20.7	24.6	25.7	24.0	21.2	19.1	14.6	12.3	(30)	
	5%	WB	9.9	9.3	10.8	12.6	15.4	18.1	18.8	18.6	17.5	15.7	12.6	10.7	(31)	
		MCDB	10.6	10.5	13.1	16.6	18.9	22.3	22.5	21.8	20.0	17.5	13.6	11.7	(32)	
	10%	WB	8.9	8.5	9.8	11.4	14.2	16.9	17.8	17.7	16.8	15.0	11.7	9.7	(33)	
		MCDB	9.7	9.7	11.7	14.4	17.3	19.8	21.0	20.3	19.0	16.6	12.8	10.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	4.6	5.6	6.8	6.5	6.9	6.4	5.9	6.3	5.2	4.1	4.0	(35)	
		MCDBR	3.8	6.2	8.6	10.7	10.8	11.8	11.6	10.3	9.6	6.5	4.5	3.9	(36)	
	5% WB	MCWBR	3.7	4.4	5.2	6.0	6.0	6.2	5.7	5.1	5.0	4.0	3.7	3.6	(37)	
		MCWBR	3.7	4.8	7.4	9.5	9.4	10.5	9.8	8.2	7.3	5.9	4.3	3.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.344	0.378	0.396	0.401	0.400	0.395	0.390	0.374	0.359	0.334	0.321	(40)	
		taud	2.431	2.408	2.313	2.283	2.313	2.346	2.363	2.398	2.426	2.456	2.451	2.400	(41)	
	Ebn at Noon		720	797	825	848	858	860	859	847	823	769	704	667	(42)	
		Edh at Noon	64	82	106	121	123	120	116	107	94	78	62	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.88	1.62	2.79	4.36	5.12	5.71	5.48	4.60	3.54	2.02	1.06	0.73	(44)	
	RadStd		0.09	0.21	0.31	0.43	0.42	0.55	0.38	0.34	0.22	0.18	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page